

Practice session 3

The following exercises refer to the topics covered in the previous lessons. Some of them will be carried out together with the teacher. Others will have to be carried out individually, and - at the end of the allotted time - the teacher will illustrate and comment on their correct execution. Exercises that will not be carried out in the classroom must in any case be considered an integral part of the Practice, to be carried out independently at home.

The Tutors present in the classroom during the Practice can only provide support in the execution of some steps. Any doubts about the approach used by the teacher and the validity of any alternative solutions should be clarified exclusively with the teacher.

Exercise 1

In a new Python file named **Exercise 1.py**, create the function **counter** that receives a list containing words as mandatory argument and returns a list of integers representing the length of each word of the first list.

Exercise 2

The file **Exercise 2.py** contains a Python function that creates two lists of random numbers and returns their common elements in a new list. The function is incomplete and must:

- receive an integer as a mandatory parameter
- create two lists of random numbers, each one of length equal to the integer received as mandatory argument
- return a new list containing all the numbers that are present in both lists of random numbers

For example, if the two lists of random numbers are [34,65,51,12] and [65,18,12,87], the function must return the list [65,12].

Exercise 3

In a new Python file named **Exercise 3.py**, create the function **sum_digits** that returns the sum of the digits of a number. In particular, the function must:

- ask the user to enter an integer number, to be stored as a string
- calculate the sum of the digits of the number entered by the user
- return the sum

For example, if the user enters the number 2154, the function must return 12 (i.e. $2 + 1 + 5 + 4$).

Test the function several times, entering different numbers each time.

Exercise 4

Using the file **Exercise 4.py** create the **string_count** function to count the number of text strings in a list. In particular, the function must:

- have a list as a mandatory parameter
- count how many text strings are contained in the list received as a mandatory argument
- return the number of text strings

To test the function you can use the list *economics* (present in the file) which consists of a series of words and numbers.

Then, create the **strings** function that:

- has a list as mandatory parameter
- returns a new list containing only the text strings found in the list passed as a mandatory argument

To test the function you can use the list *economics*

Finally, create the **analysis** function that:

- has a list as a mandatory parameter
- counts how many text strings are contained in the list received as a mandatory argument
- sums all the numeric elements of the list passed as a mandatory argument
- returns a new list containing two elements, i.e. the count of text strings and the sum of numeric elements found in the list received as mandatory argument

To test the function you can use the list *economics*

Exercise 5

In a new Python file named **Exercise 5.py**, create the function **character_count** to count the number of occurrences of a character within a text string passed as an argument. The function must:

- have a text string passed as a mandatory argument
- have the character to be found within the text string as an optional parameter (the default value should be "a")
- count how many times the character is repeated within the text string
- return the number of occurrences of the character in the text string